

Exam Date & Time: 26-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - Make up Examination - December -2024

HUMAN COMPUTER INTERFACE [CSE-4069]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) What is the goal and importance of human computer interface. Why human computer interface development is multidisciplinary? Discuss. (5)
 - A)
 - B) Which input and output devices would you select for the following systems?
 - i) tractor-mounted crop-spraying controller
 - ii) air traffic control system(5)
For each device, evaluate the alternatives and highlight any relevant reasons why traditional options like the keyboard, mouse, and CRT monitor might be less ideal.
- 2) What is WIMP interface? How different elements of WIMP can be designed for tourist information systems? Discuss. (5)
 - A)
 - B) What is the need and application of the following computing paradigms? Explain.
(a) Time Sharing (b) Ubiquitous computing (5)
- 3) Explain in detail Shneiderman's golden rules for effective design of human computer interface (5)
 - A)
 - B) What is the need of iterative design and prototyping? Describe with a neat diagram three different prototypes that can be applied in software development. (5)
- 4) Compare and contrast HCI evaluation through User and Expert analysis. Give one example for each. (5)
 - A)
 - B) What is the need of user support in HCI systems? How user support can be provided for effective usage of HCI systems? Give any two examples. (5)
- 5) Why is it important to consider different stakeholders in HCI design process? Who are the different stakeholders in the airline booking system?. Categorize them into different (5)

levels and provide proper reasons for categorization.

A)

B) Suppose a project team has to develop an online meeting platform for group discussions for an organization. How does this new platform affect the communication of the employees in office meetings? (5)

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